



اَبُو سَیِّدِی تَیْکُو لُوْجِی مَیْاَرَا
UNIVERSITI
TEKNOLOGI
MARA

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Sains Komputer
dan Matematik

FACULTY OF COMPUTER AND MATHEMATICAL SCIENCES
BACHELOR OF COMPUTER SCIENCE (HONS.)

CSC584
ENTERPRISE PROGRAMMING

GROUP PROJECT:

SpeakOut - Anonymized Bullying Reporting and Support Platform

GROUP:

NBCS2306A - GROUP 3

SEMESTER MARCH 2026 – AUGUST 2026

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SUBMISSION DATE:

31 MAY 2026

SCORESHEET PROPOSAL

			Comments
Description			
Overview, problem statement, objective	/ 5		
Design/storyboard/template	/ 5		
Total 1		/ 10	

			Comments
Database Design			
Table structure	/ 3		
Relationship between tables	/ 3		
Total 2		/ 6	

Flow of Application			
Create/Read	/ 1		
Update	/ 1		
Delete	/ 1		
Report (Analysis)	/ 1		
Total 3		/ 4	

Grand Total	/ 20
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SCORESHEET INFORMATION SYSTEM

			Comments
Input Validation and Security			
Input validation	/ 2		
Session management	/ 2		
Total 1	/ 4		

System Functionalities			
Create/Read (with success message)	/ 2		
Update (with confirmation and success message)	/ 2		
Delete (with confirmation and success message)	/ 2		
Report Generation	/ 2		
Total 2	/ 8		

User Interface			
Screen design (use template like Bootstrap, TailwindCSS, etc)	/ 4		
User friendly (consistent navigation bar/menu on every screen)	/ 4		
Total 3	/ 8		

SCORESHEET GROUP PROJECT PRESENTATION

Video Presentation			
Content organization	/ 4		
Use of media (image, sound, etc)	/ 2		
Oral presentation skill (clear voice, etc)	/ 4		
Total 5	/ 10		

Grand Total	/ 30
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1.0 PROJECT OVERVIEW

SpeakOut is a secure web-based enterprise information system that allows students and the school community to report a bullying incident anonymously, with a unique Case ID that can be tracked from a multi-client Python and analytics prototype in CSC577 to a single, standards-based, Java web application for the CSC584 Enterprise Programming requirements.

The platform has been developed with a solid MVC (Model-View-Controller) design, and utilizes a combination of JavaBeans (Model), Servlets (Controller), and JSP (View) technology along with JSTL and Expression Language to separate the business logic from scriptlets, enabling multi-school multi-tenancy (one deployment, multiple schools). It uses secure session-based authentication, server side registration validation, full CRUD operations for persisted case data and role-aware dashboards styled with bootstrap 5 and powered by Chart.js for administrators to compare regional KPIs and outcomes.

In addition, the system boosts incident response and management speeds by allowing students to submit evidence (e.g pictures, videos, documents) to a comprehensive dashboard that can be discussed, prioritised, identified and solved by set school authorities, who are using a structured branch/pull-request workflow in GitHub to build its features.

2.0 PROBLEM STATEMENT

Bullying is a universal problem in schools and for many reasons, it is not reported due to the following issues:

- **Fear of retaliation:** Victim or bystanders hesitate to report as they may be afraid of social consequences and/or fear that the situation will get worse if it is reported.
- **Lack of anonymous reporting channels:** Most schools use manual, verbal or paper-based reporting systems, which do not ensure confidentiality and students will not come forward.
- **Poor case tracking:** Paper forms and spreadsheets can lead to missed data, poor record keeping, and transparency issues for follow-up actions.
- **Distrust in the system:** If students don't see meaningful and timely responses to previous reports, they lose trust in the system, and this soon diminishes their motivation to report future incidents.
- **Underawareness:** People do not know their rights and the ways to report, thus further underreported.

The problems taken together make it a climate conducive to the continuation of bullying, which has a negative impact on the well-being, performance and psychological development of students. It is therefore imperative that there is a system of reporting in place that is structured, digital and confidential, establishes trust, allows for timely responses and gives school authorities the tools they need to make effective decisions based on data.

3.0 OBJECTIVE

The following objectives have been set for SpeakOut system based on above identified problems:

- **Secure Authentication & Role-Aware Authorization:** To create a Login/Logout module with secure session management ensuring protected access with strict role-aware authorization depending on the users (Student, Teacher/Case Handler, Administrator, PIBG, PDRM).
- **Validated & Secure Registration:** To create a complete Registration module with validated input on the client side and rules on the server side to validate the input (e.g. email formatting, password strength, required fields, etc., and credential uniqueness).
- **School-Based Multi-Tenant Information Management (CRUD):** To develop a complete implementation of Information Management module with school level multi-tenancy at the DAO and Servlet level. The module will enable secure Create, Read, Update, and Delete (CRUD) of bullying incident reports and teacher case notes stored on a relational database of at least four tables, including enhanced search/filter functionality, structural transitions for the status of the incidents and confirmation workflows for destructive operations.
- **Role-Specific Analytical Dashboards:** An interactive Dashboard module displaying unique operational insights and data visualizations for each user role, through KPI cards, trend charts, graphs, heatmaps and recent activity lists, etc. to track case statuses and bullying trends.
- **Standardized Java MVC Architecture:** To design and implement the web application with the Java technology (MVC framework) which includes java bean, JSP and Servlet components with high structured, modular and maintainable code base.
- **Collaborative Version Control:** To use **GitHub** as the primary version control tools to provide a seamless way to collaborate with team members, merge codes and track the development process of software.
- **Comprehensive Project Deliverables:** A complete HTML prototype (storyboard), a fully functioning web application that meets all the architectural levels, and a detailed user manual will be produced and delivered.

4.0 PROPOSED USER INTERFACE

The following proposed user interfaces show the main screens and the flow of navigation through the SpeakOut web application. All the screens are built with Bootstrap to ensure a consistent, responsive and user friendly experience.

4.1 Login / Logout Screen

The Logon screen enables users to log on with their credentials in a secure manner, where users include registered users such as students, teachers, administrators, PIBG, and PDRM. A session is created upon successful login and destroyed upon logout. Input validation, error message and a consistent navigation bar on each authenticated page.

SpeakOut
Anonymous Bullying Reporting & Support Platform

A safer space to speak up.

Anonymous bullying reporting for students.
Transparent case tracking for teachers.

"Bila ada platform macam ni, lagi senang nak report tanpa rasa takut."
- Student, SMK Seksyen 9

Identity hidden from peers - visible only to your school's counsellor.

Welcome Back

Login to continue to SpeakOut.

Email Address

Enter your email

Password [Forgot Password?](#)

Enter your password

☐ Keep me logged in on this device

Login to Your Space

or quick-preview as

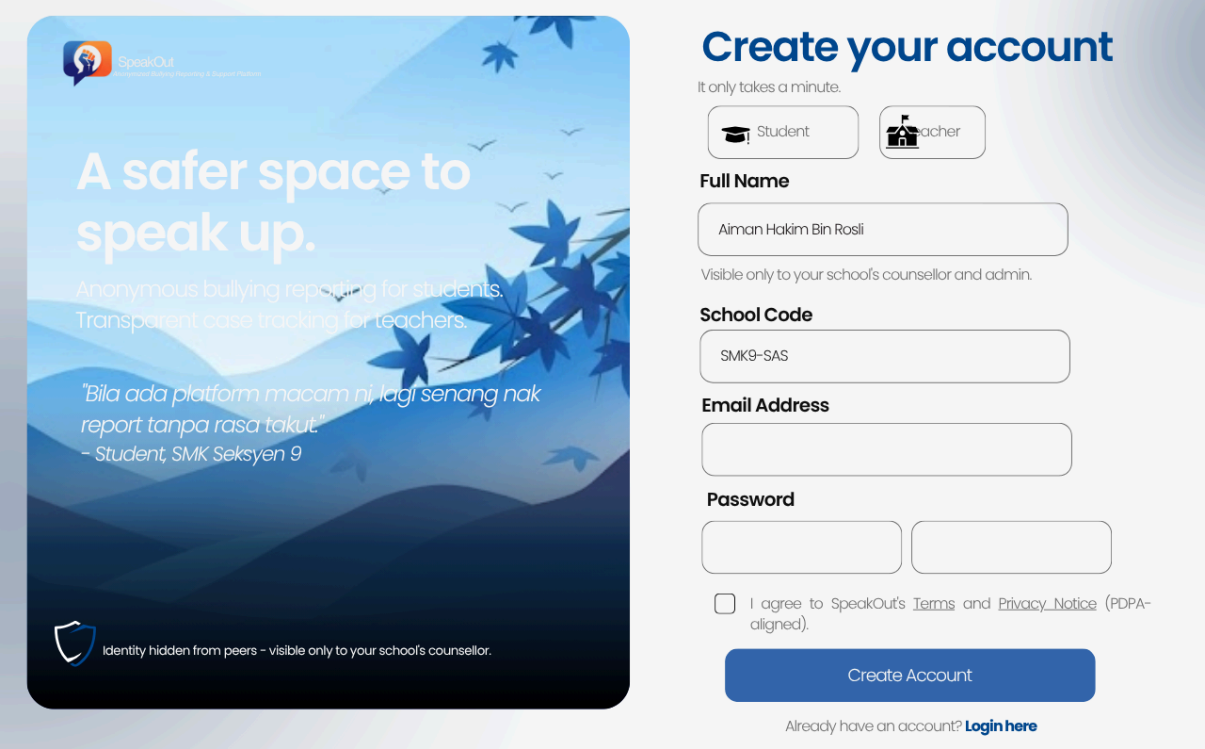
Student Teacher Admin

Don't have an account? [Register here](#)

[WEB Login Screen Mockup]

4.2 Registration Screen

If you are a new user, you can register on the Registration screen. It will have input validity for all input (name, email, password strength and role selection). The system checks before an account is created that all the necessary fields have been filled out, and that the email is not already in use. Errors are presented to the user and will prompt them to correct their input.



The mockup shows a registration interface for 'SpeakOut'. On the left is a promotional card with a blue background and white text. It features the 'SpeakOut' logo, the headline 'A safer space to speak up.', and a quote from a student. On the right is the 'Create your account' form. It includes role selection buttons for 'Student' and 'Teacher', followed by input fields for 'Full Name', 'School Code', 'Email Address', and 'Password'. A checkbox for terms and privacy is present, along with a 'Create Account' button and a 'Login here' link.

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- Student, SMK Seksyen 9

Identity hidden from peers - visible only to your school's counsellor.

Create your account

It only takes a minute.

☐ Student ☐ Teacher

Full Name
Aiman Hakim Bin Rosli
Visible only to your school's counsellor and admin.

School Code
SMK9-SAS

Email Address

Password

☐ I agree to SpeakOut's [Terms](#) and [Privacy Notice](#) (PDPA-aligned).

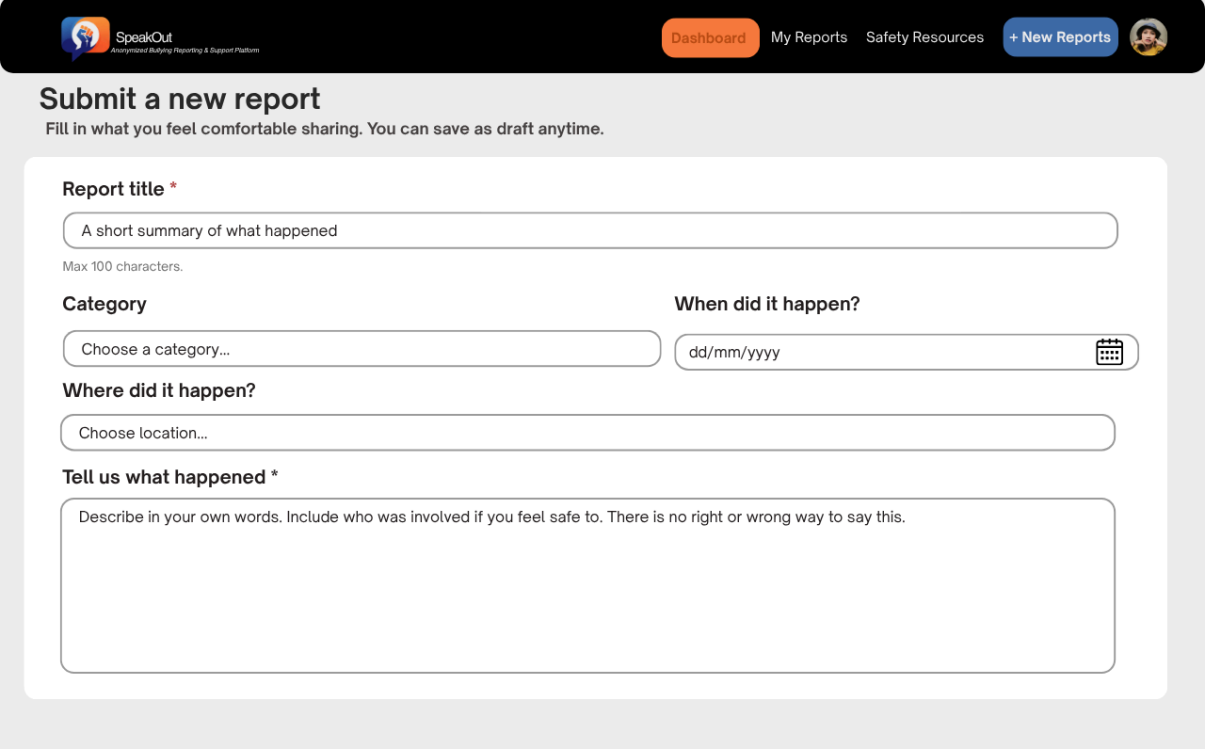
Create Account

Already have an account? [Login here](#)

[WEB Registration Screen Mockup]

4.3 Report Submission Screen (Create)

The Report Submission screen gives students an opportunity to submit a new report of a bullying situation anonymously. Users will be able to outline the situation, pick from a list of bullying types, upload supporting evidence (images/videos/documents up to 50MB) and submit the report. A Case ID is created and shown to the reporter upon submission so that he can monitor it later.



The mockup shows a web interface for submitting a report. At the top is a dark navigation bar with the 'SpeakOut' logo and text 'Anonymous Bullying Reporting & Support Platform'. Navigation links include 'Dashboard' (highlighted in orange), 'My Reports', 'Safety Resources', and '+ New Reports' (highlighted in blue) next to a user profile icon. The main heading is 'Submit a new report' with a subtext: 'Fill in what you feel comfortable sharing. You can save as draft anytime.' The form fields are: 'Report title *' with a placeholder 'A short summary of what happened' and a 'Max 100 characters.' note; 'Category' with a dropdown 'Choose a category...'; 'When did it happen?' with a date input 'dd/mm/yyyy' and a calendar icon; 'Where did it happen?' with a dropdown 'Choose location...'; and 'Tell us what happened *' with a large text area and placeholder text: 'Describe in your own words. Include who was involved if you feel safe to. There is no right or wrong way to say this.'

[Report Submission Screen Mockup]

4.4 Case Management Dashboard (Read / Update / Delete)

Teachers, administrators and authorised authorities (PDRM, PIBG) can access the Case Management Dashboard. It presents all reported cases in a tabular format and has filtering and search functions. Users can Read, Update the case status, add Investigation notes or delete invalid/resolved records for the case. There is a confirmation dialog box for each action and a success message for each successful action.

The mockup shows a user interface for the 'SpeakOut' platform. At the top is a navigation bar with the logo, 'Dashboard', 'My Reports', 'Safety Resources', and a '+ New Reports' button. The main content area displays a case titled 'Repeated name-calling in chemistry class'. It includes a status bar with 'In review' (yellow), 'Verbal' (pink), and 'Submitted 2025-05-15 Classroom'. Below this is a 'What I reported' section with a text description of the incident. To the right, the 'Assigned counsellor' is 'Cikgu Siti Aminah'. A 'Case details' sidebar lists attributes like Case ID (C#SO-0421), Status (In review), Category (Verbal), Location (Classroom), Incident date (2025-05-15), Submitted time (2025-05-15 09:42), and Anonymity (Hidden from peers). A 'Case timeline' section at the bottom left shows three events: report submission, assignment to the counsellor, and a status change to 'In review'.

 SpeakOut
Anonymized Bullying Reporting & Support Platform

DashboardMy ReportsSafety Resources+ New Reports

Repeated name-calling in chemistry class

Fill in what you feel comfortable sharing. You can save as draft anytime.

In reviewVerbal

Submitted 2025-05-15Classroom

Case ID #SO-0421

Assigned counsellor

 **Cikgu Siti Aminah**
Counsellor, SMK Seksyen 9

What I reported

For the past two weeks, a group of three students in my chemistry class have been calling me names and laughing at me whenever I answer questions. Today they pretended to "sneeze" out my nickname loudly while Cikgu Hasan had his back turned. I tried to ignore it but it's getting harder to focus.

Case details

Case ID	C#SO-0421
Status	In review
Category	Verbal
Location	Classroom
Incident date	2025-05-15
Submitted	2025-05-15 09:42
Anonymity	Hidden from peers

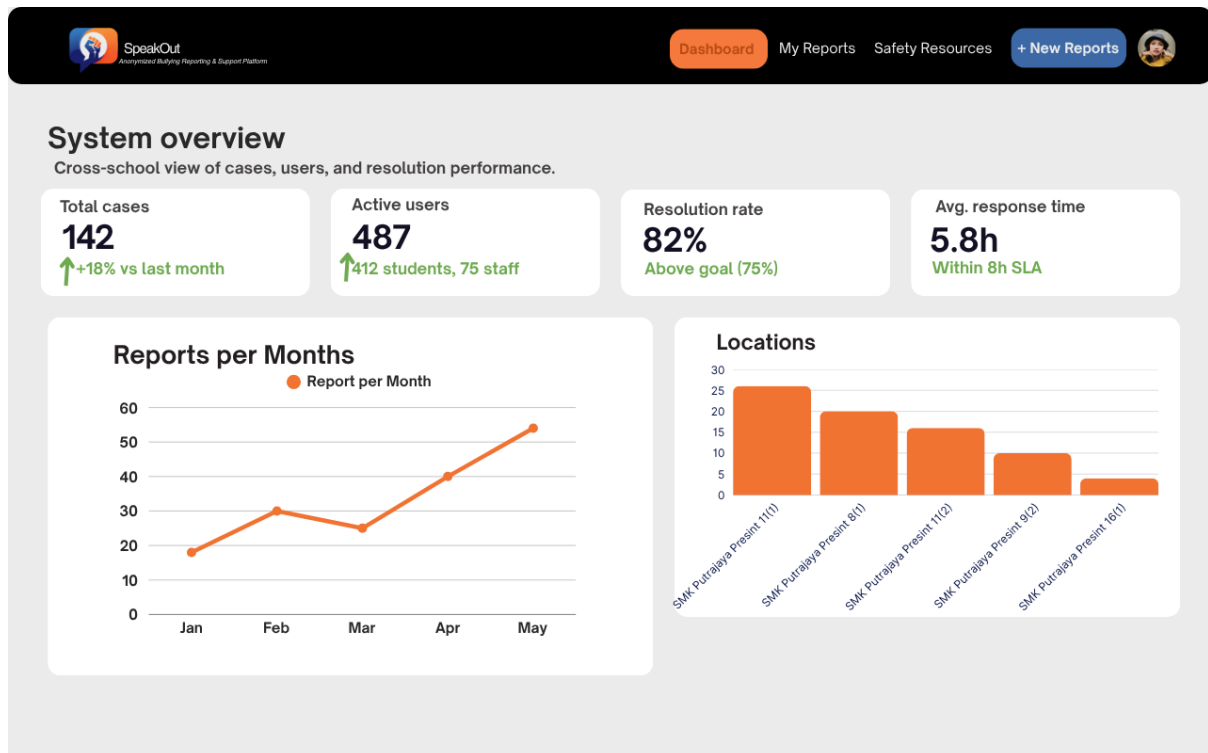
Case timeline

- 2025-05-15 09:42
Report submitted
Status set to Submitted. Routed to counsellor.
- 2025-05-15 10:18
Assigned to Cikgu Siti Aminah
School counsellor.
- 2025-05-15 11:55
Status changed: Submitted → In review
Counsellor began reviewing the report.

[Case Management Dashboard Mockup]

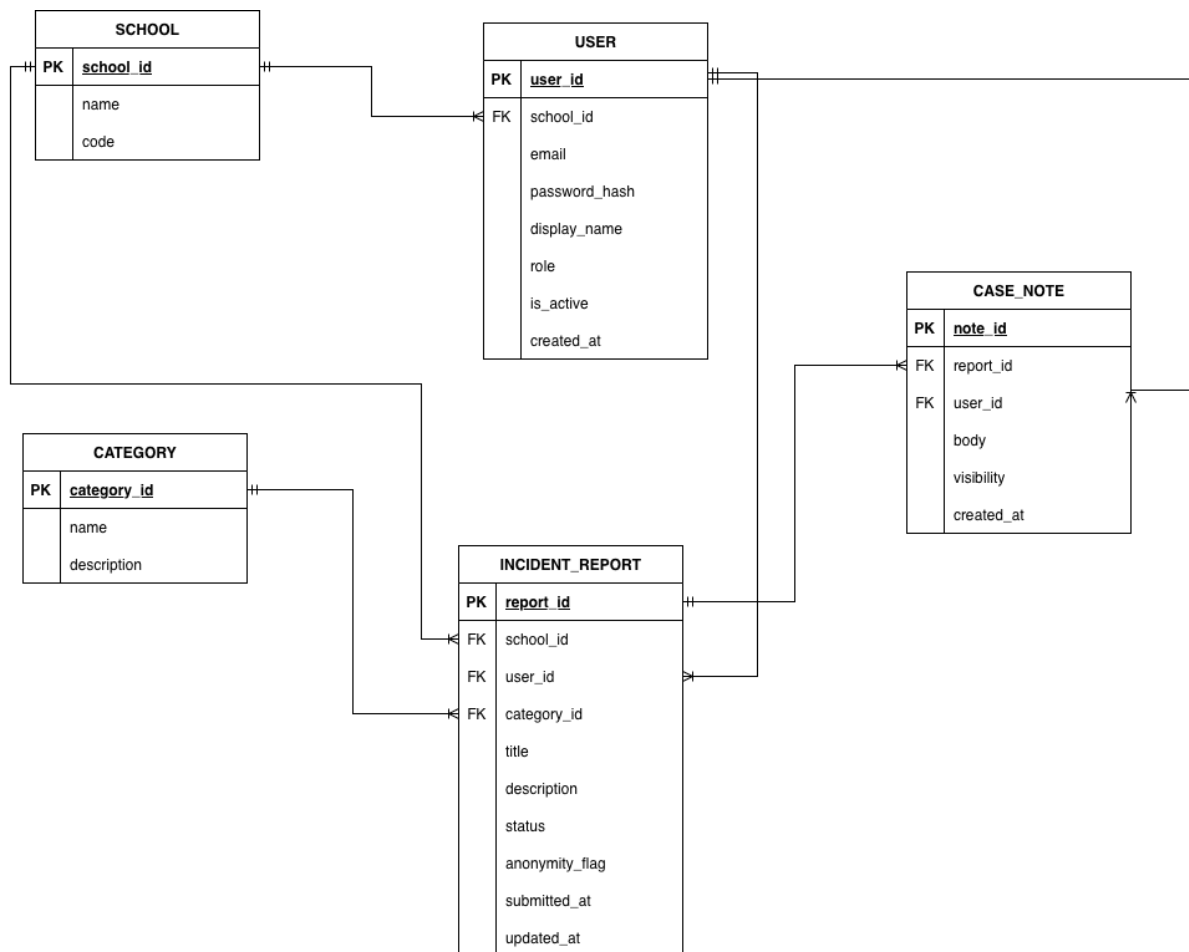
4.5 Analytics Dashboard (Report / Overview)

Admin and authorized users can use the Analytics Dashboard to get a clear, visual view of bullying trends, case status and response metrics. It contains bar charts, pie charts and trend line graphs which are dynamically updated when the data held in the database is changed. This screen meets the requirement for the Dashboard module and the requirement to provide a report (Analysis) for the project.



[Analytics Dashboard Mockup]

5.0 DATABASE DESIGN



Business Rules

- **School (1:M) User**: each user has a single school that is the basis for multi-school reporting.
- **User (1:M) Incident_report (as student)**: students author many reports.
- **School (1:M) Incident_report**: applies tenant boundary to queries and dashboards.
- **Category (1:M) Incident_report**: normalised category dimension for analytics.
- **Incident_report (1:M) Case_note**: teachers add structured notes, supports CRUD on notes and status change to the parent report.

5.1 Tables of Entity-Relationship

The SpeakOut database is designed in a relational model, which meets the project requirement, with at least four tables. The following Entity-Relationship (ER) diagram shows the entities, their attributes and their relationship (with the data type).

Table 1: USERS

Column Name	Data Type	Constraint	Description
user_id	VARCHAR(50)	PRIMARY KEY	Unique identifier
full_name	VARCHAR(100)	NOT NULL	User's full name
email	VARCHAR(100)	NOT NULL, UNIQUE	Login email address
password_hash	VARCHAR(255)	NOT NULL	Bcrypt-hashed password
role	ENUM	NOT NULL	Student / Teacher / Admin / PIBG / PDRM
is_active	BOOLEAN	DEFAULT TRUE	Account active status
created_at	DATETIME	DEFAULT NOW()	Account creation timestamp

Table 2: REPORTS

Column Name	Data Type	Constraint	Description
report_id	VARCHAR(50)	PRIMARY KEY	Unique report identifier
reporter_id	VARCHAR(50)	FOREIGN KEY → USERS	Anonymous reporter reference
description	TEXT	NOT NULL	Incident description text
bullying_type	ENUM	NOT NULL	Cyber / Verbal / Physical / Social
severity	ENUM	NOT NULL	Low / Medium / High / Critical
keywords	TEXT	NULL	TF-IDF extracted keywords
submitted_at	DATETIME	DEFAULT NOW()	Submission timestamp

Table 3: CASES

Column Name	Data Type	Constraint	Description
case_id	VARCHAR(50)	PRIMARY KEY	Unique case identifier (e.g. SO-2026-00001)
report_id	VARCHAR(50)	FOREIGN KEY → REPORTS	Linked report
assigned_to	VARCHAR(50)	FOREIGN KEY → USERS	Assigned authority user
status	ENUM	NOT NULL	New / Under Investigation / Resolved / Closed
priority	ENUM	NOT NULL	Low / Medium / High / Critical
resolution_notes	TEXT	NULL	Investigation and resolution notes
created_at	DATETIME	DEFAULT NOW()	Case creation timestamp
updated_at	DATETIME	DEFAULT NOW()	Last update timestamp

Table 4: EVIDENCE

Column Name	Data Type	Constraint	Description
evidence_id	VARCHAR(50)	PRIMARY KEY	Unique evidence record ID
report_id	VARCHAR(50)	FOREIGN KEY → REPORTS	Linked report
file_url	VARCHAR(500)	NOT NULL	URL of stored evidence file
file_type	VARCHAR(20)	NOT NULL	image / video / document
file_size	INT	NOT NULL	File size in bytes (max 50MB)
uploaded_at	DATETIME	DEFAULT NOW()	Upload timestamp

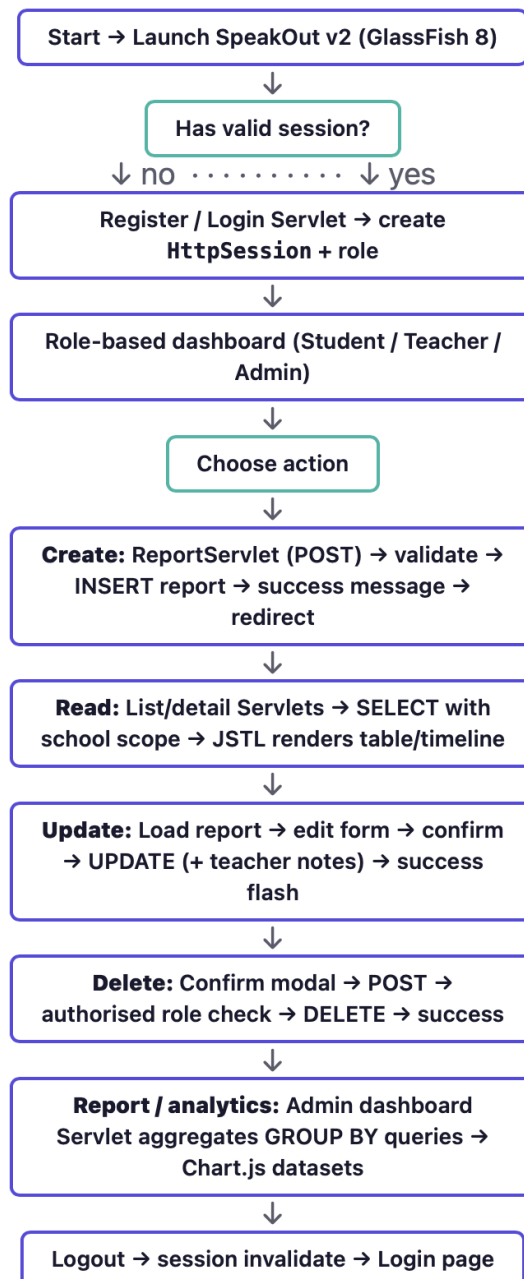
5.2 Entity-Relationship Overview

Relationship	Cardinality	Description
USERS → REPORTS	One-to-Many	One user (student) can submit multiple reports
REPORTS → CASES	One-to-One	Each report generates exactly one case record
REPORTS → EVIDENCE	One-to-Many	One report can have multiple evidence files
USERS → CASES	One-to-Many	One authority (teacher/admin) can be assigned multiple cases

6.0 FLOW OF APPLICATION

The following flowchart shows end-to-end usage for Create, Read, Update, Delete and analytic reporting on the Dashboard from the administrator. Servlets check/validate data, call DAOs, set request/session attributes, and pass on to JSP views.

It includes all 4 mandatory modules: Login/Logout, Registration, Dashboard and Information Management (CRUD). The flowchart below describes the end-to-end interaction from user entry to case resolution.



6.1 Student / Reporter Flow

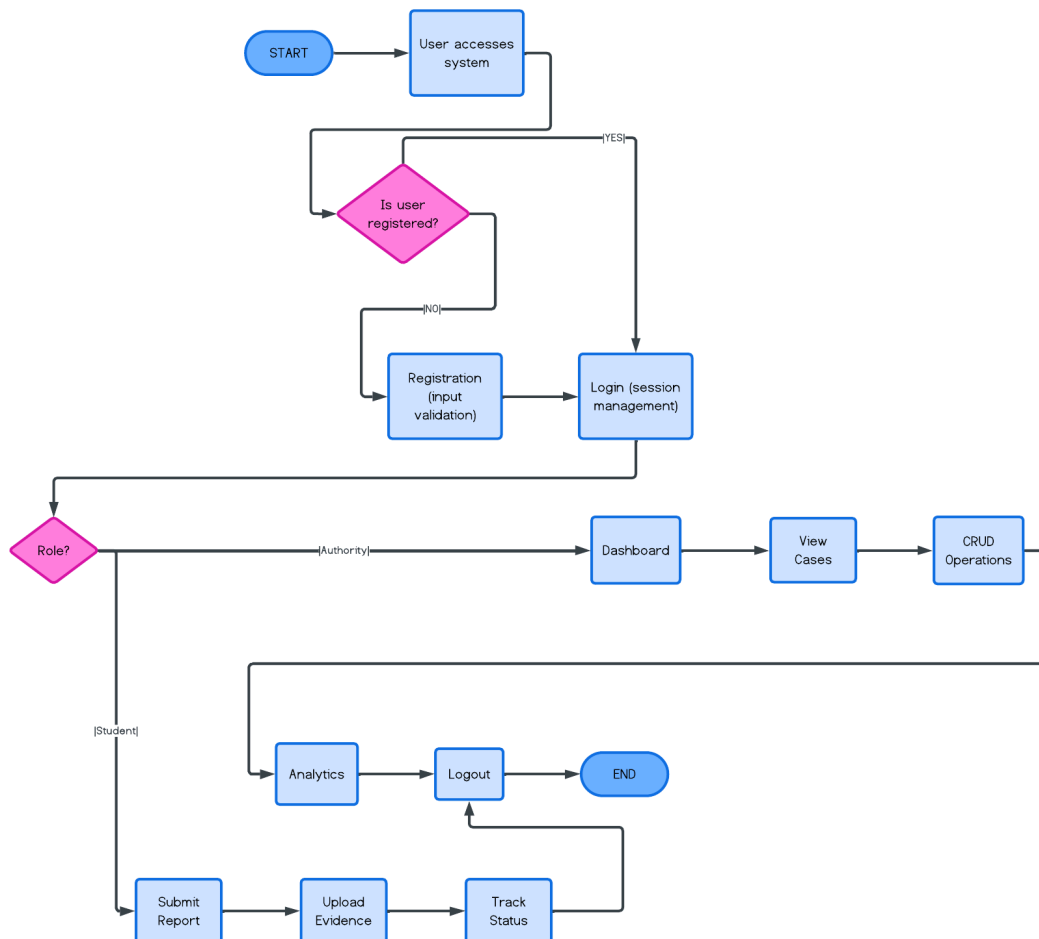
Step	Action	System Response
1	Access SpeakOut URL	Display Login / Registration page
2	Register new account (with validation)	Validate inputs → Create user record → Redirect to Login
3	Login with credentials	Authenticate → Create session → Show Student Dashboard
4	Submit bullying report (Create)	Validate form → Save to REPORTS table → Generate Case ID → Show success message
5	Upload evidence files	Validate file (type, size < 50MB) → Save to EVIDENCE table → Link to report
6	Track case status (Read)	Query CASES table by Case ID → Display current status
7	Logout	Destroy session → Redirect to Login page

6.2 Authority (Teacher / Admin) Flow

Step	Action	System Response
1	Login with authority credentials	Authenticate → Create session → Show Authority Dashboard with analytics overview
2	View Dashboard (Analytics)	Display charts: case status, bullying type distribution, trends over time
3	Read all reported cases	Query CASES + REPORTS tables → Display case list with filters
4	View case details (Read)	Display full report details, evidence files, investigation notes
5	Update case status and notes (Update)	Confirm action → Update CASES table → Show success message → Notify reporter
6	Delete invalid/spam record (Delete)	Confirmation dialog → Delete from CASES/REPORTS → Show success message

7	Generate analytics report	Aggregate data from database → Display charts, heatmaps, trend graphs
8	Logout	Destroy session → Redirect to Login page

6.3 Application Flowchart Description - User



6.4 System Technology Specification

Component	Technology / Tool
Frontend (View)	JSP (JavaServer Pages) + Bootstrap / TailwindCSS
Backend (Controller)	Java Servlet (MVC Framework)
Model (Business Logic)	JavaBean
Database	MySQL / PostgreSQL (min. 4 tables)
Version Control	GitHub (collaborative repository)
Session Management	Java HttpSession for Login/Logout
Input Validation	Server-side (Servlet) + Client-side (JavaScript)